MDM Study Guides: all knowledge areas

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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1.A Engineering Terms, Symbols, and Drawings

1. Key concepts

- Orthographic views: US practice is third-angle projection. The top view sits above the front view and the right-side view sits to its right. First-angle (ISO-E) mirrors that layout.
- Section views show internal features. Full section: the cutting plane goes all the way through and the cut surface is hatched. Half section: one quarter is removed. Broken-out, revolved and removed sections are local variants.
- Dimension notation: a boxed dimension is BASIC (theoretically exact; the tolerance comes from a feature control frame). A dimension in parentheses is REFERENCE (information only, not inspected).
- GD&T (ASME Y14.5) has five families. Form (flatness, straightness, circularity, cylindricity) never takes a datum. Orientation (perpendicularity, parallelism, angularity), location (position, concentricity, symmetry) and runout (circular, total) need datums. Profile can go either way.
- Material conditions: MMC is the condition with the most material (smallest hole, largest shaft). LMC is the opposite. RFS is the default. A bonus tolerance applies when a feature departs from MMC and the MMC modifier is used.
- Virtual condition (VC) is the worst-case boundary. Shaft: VC = MMC + geometric tolerance. Hole: VC = MMC − geometric tolerance. VC sets the size of functional gauges and the mating clearance.

2. Essential formulas

Name	Formula	Variables / units
Bonus tolerance	$\text{Total position tol} = \text{stated tol} + \lvert\text{actual size} - \text{MMC size}\rvert$	valid only with the Ⓜ modifier; sizes in in or mm
Virtual condition (external feature)	$\text{VC} = \text{MMC} + \text{geometric tol}$	shaft/pin
Virtual condition (internal feature)	$\text{VC} = \text{MMC} - \text{geometric tol}$	hole
Position from coordinates	$\text{Diametral position error} = 2\sqrt{(\Delta x^2 + \Delta y^2)}$	Δx, Δy = measured deviation from the true position

3. Handbook search keywords

third angle projection · section view · GD&T · position tolerance · maximum material condition · virtual condition · datum

4. Worked examples

Example 1: Bonus tolerance and position check
A Ø0.500 +0.006/−0.000 in hole has position tolerance Ø0.010 at MMC. A hole measures Ø0.504, and its axis is 0.003 in off in x and 0.004 in off in y. Is it acceptable?

1. MMC of the hole = smallest size = 0.500 in.
2. Bonus = 0.504 − 0.500 = 0.004 in, so the allowed position = 0.010 + 0.004 = 0.014 in (diameter zone).
3. Actual diametral error = $2\sqrt{(0.003^2 + 0.004^2)} = 0.010$ in.
4. $0.010 \le 0.014$, so the hole is acceptable.

Answer: Acceptable (0.010 in used of 0.014 in allowed)

Example 2: Virtual condition of a pin
A pin is Ø0.750 ±0.002 in with perpendicularity Ø0.003 at MMC. Find its virtual condition.

1. MMC of the pin = largest size = 0.752 in.
2. VC = MMC + tol = 0.752 + 0.003 = 0.755 in.

Answer: 0.755 in

5. Common traps

- Forgetting that the MMC of a HOLE is its SMALLEST size.
- Using the radial deviation $\sqrt{(\Delta x^2+\Delta y^2)}$ instead of the diametral $2\sqrt{(\Delta x^2+\Delta y^2)}$.
- Adding a bonus tolerance when the frame has no Ⓜ modifier (RFS gives no bonus).
- Mixing up first-angle and third-angle view placement.

1.A Engineering Terms, Symbols, and Drawings: one-page cheat summary

- ▶ 3rd angle: top view above, right view to the right
- ▶ Boxed = basic; (x) = reference
- ▶ Form tolerances take no datum
- ▶ Hole MMC = smallest; shaft MMC = largest
- ▶ Bonus = |actual − MMC| (Ⓜ only)
- ▶ Position error = $2\sqrt{(\Delta x^2+\Delta y^2)}$
- ▶ VC shaft = MMC + tol; VC hole = MMC − tol

Formulas

Bonus tolerance	Total position tol = stated tol + |actual size − MMC size|
Virtual condition (external feature)	VC = MMC + geometric tol
Virtual condition (internal feature)	VC = MMC − geometric tol
Position from coordinates	Diametral position error = $2\sqrt{(\Delta x^2 + \Delta y^2)}$

Search keywords

third angle projection, section view, GD&T, position tolerance, maximum material condition, virtual condition, datum

Top traps

- Forgetting that the MMC of a HOLE is its SMALLEST size.
- Using the radial deviation $\sqrt{(\Delta x^2+\Delta y^2)}$ instead of the diametral $2\sqrt{(\Delta x^2+\Delta y^2)}$.
- Adding a bonus tolerance when the frame has no Ⓜ modifier (RFS gives no bonus).

1.B Project Management and Economic Analysis

1. Key concepts

- Time value of money: move every cash flow to the same point in time (P, F or A) at the interest rate i before you compare.
- Mutually exclusive alternatives: pick the best PW, AW or EUAC. With unequal lives, use annual worth, or PW over the least common multiple of the lives.
- Sunk costs are irrelevant. Only future cash flows that differ between alternatives count.
- Break-even: the volume at which revenue equals total cost. Payback ignores the time value of money and anything after payback.
- Depreciation: straight line D = (P − S)/n. MACRS uses table percentages. Book value = cost − accumulated depreciation.
- Project scheduling (CPM): the critical path is the LONGEST path through the network, and its activities have zero total float. Earned value: CPI = EV/AC, SPI = EV/PV (above 1 is good).

2. Essential formulas

Name	Formula	Variables / units
Single payment	$F = P(1+i)^{n}$; $P = F(1+i)^{-n}$	i per period, n periods
Uniform series PW	$P = A[(1+i)^{n}-1]/[i(1+i)^{n}]$	A at the end of each period
Capital recovery	$A = P{\cdot}i(1+i)^{n}/[(1+i)^{n}-1]$	(A/P,i,n)
Sinking fund	$A = F{\cdot}i/[(1+i)^{n}-1]$	(A/F,i,n)
EUAC	$\mathrm{EUAC} = P(A/P) - S(A/F) + \text{annual O\&M}$	S = salvage
Break-even	$N = \text{Fixed cost}/(\text{price} - \text{variable cost per unit})$	
Effective rate	i_eff = (1 + r/m)^m − 1	r nominal, m compoundings/yr
Earned value	$\mathrm{CPI} = \mathrm{EV}/\mathrm{AC}$; $\mathrm{SPI} = \mathrm{EV}/\mathrm{PV}$	

3. Handbook search keywords

interest factor tables · capital recovery · sinking fund · equivalent uniform annual cost · break-even · depreciation · critical path

4. Worked examples

Example 1: EUAC of a machine
A machine costs $50,000, lasts 10 years, has a $8,000 salvage value and $3,000/yr O&M. At i = 8%, find its EUAC.

1. (A/P,8%,10) = 0.14903; (A/F,8%,10) = 0.06903.
2. EUAC = 50,000(0.14903) − 8,000(0.06903) + 3,000 = $9,899/yr.

Answer: ≈ $9,899/yr

Example 2: Break-even volume
Fixed cost $90,000/yr, price $45/unit, variable cost $27/unit. Find the break-even volume.

1. N = 90,000/(45 − 27) = 5,000 units/yr.

Answer: 5,000 units/yr

5. Common traps

- Using a nominal rate as if it were the effective rate when compounding is monthly.
- Treating salvage as a cost (it is a credit, so subtract it).
- Comparing PW of alternatives with unequal lives over their own lives.
- Including sunk costs in a replacement study.
- Calling the path with the most activities the critical path (it is the longest in time).

1.B Project Management and Economic Analysis: one-page cheat summary

- ► $(A/P) = i(1+i)^n/[(1+i)^n-1]$
- ► $(A/F) = i/[(1+i)^n-1] = (A/P) - i$
- ► $(P/A) = 1/(A/P)$
- ► $EUAC = P(A/P) - S(A/F) + O\&M$
- ► Break-even $N = FC/(p - v)$
- ► SL depreciation $(P-S)/n$
- ► Critical path = longest, zero float

Formulas

Single payment	$F = P(1+i)^n$; $P = F(1+i)^{-n}$
Uniform series PW	$P = A[(1+i)^n-1]/[i(1+i)^n]$
Capital recovery	$A = P{\cdot}i(1+i)^n/[(1+i)^n-1]$
Sinking fund	$A = F{\cdot}i/[(1+i)^n-1]$
EUAC	EUAC = P(A/P) − S(A/F) + annual O&M
Break-even	N = Fixed cost/(price − variable cost per unit)
Effective rate	i_eff = (1 + r/m)^m − 1
Earned value	CPI = EV/AC ; SPI = EV/PV

Search keywords

interest factor tables, capital recovery, sinking fund, equivalent uniform annual cost, break-even, depreciation, critical path

Top traps

- Using a nominal rate as if it were the effective rate when compounding is monthly.
- Treating salvage as a cost (it is a credit, so subtract it).
- Comparing PW of alternatives with unequal lives over their own lives.

1.C Design Methodology

1. Key concepts

- The design process runs: define requirements → generate concepts → analyze and select (decision matrix) → detail design → prototype and test → release. It iterates.
- Verification asks 'did we build it right?' (does it meet the spec). Validation asks 'did we build the right thing?' (does it meet the user's need).
- FMEA rates each failure mode for Severity, Occurrence and Detection (1–10). RPN = S×O×D ranks the actions. A high severity alone also gets attention.
- Factor of safety n = strength/stress (or capacity/load). Reliability-based design uses distributions instead of a single n.
- Reliability: series $R = \Pi R_i$; parallel (redundant) $R = 1 - \Pi(1 - R_i)$. Constant failure rate: $R(t) = e^{(-\lambda t)}$, $MTBF = 1/\lambda$.
- Root-cause failure analysis: preserve the evidence → fractography (beach marks show fatigue, a cup-cone shows ductile overload) → root cause → corrective action → verify.

2. Essential formulas

Name	Formula	Variables / units
Factor of safety	$n = S/\sigma$ (or allowable load/actual load)	
Series reliability	$R_s = R_1 R_2 \ldots R_n$	
Parallel reliability	$R_p = 1 - (1-R_1)(1-R_2)\ldots$	
Exponential reliability	$R(t) = e^{(-\lambda t)}$; $MTBF = 1/\lambda$	λ in failures/h
FMEA	$RPN = S \times O \times D$	each 1–10

3. Handbook search keywords

factor of safety · reliability series parallel · FMEA · failure rate · design process

4. Worked examples

Example 1: System reliability
Pump A (R = 0.95) feeds two redundant valves in parallel (each R = 0.90), then a controller (R = 0.98). Find the system reliability.

1. Valve pair: $1 - (0.10)^2 = 0.99$.
2. System: $0.95 \times 0.99 \times 0.98 = 0.9217$.

Answer: 0.922

Example 2: Exponential life
A component has $\lambda = 2 \times 10^{-5}$ failures/h. Find R at 10,000 h and the MTBF.

1. $R = e^{(-0.2)} = 0.8187$.
2. $MTBF = 1/\lambda = 50{,}000$ h.

Answer: $R \approx 0.819$; MTBF = 50,000 h

5. Common traps

- Adding reliabilities, or multiplying them for a parallel set.
- Swapping verification and validation.
- Thinking a high RPN is the only trigger (a severity of 9–10 is always critical).

1.C Design Methodology: one-page cheat summary

- ► Series: multiply R
- ► Parallel: 1 − Π(1−R)
- ► R = e^(−λt), MTBF = 1/λ
- ► RPN = S·O·D
- ► n = strength/stress
- ► Verify = spec; validate = user need

Formulas

Factor of safety	$n = S/\sigma$ (or allowable load/actual load)
Series reliability	$R_s = R_1 R_2 \ldots R_n$
Parallel reliability	$R_p = 1 - (1-R_1)(1-R_2)\ldots$
Exponential reliability	$R(t) = e^{(-\lambda t)}$; $MTBF = 1/\lambda$
FMEA	$RPN = S \times O \times D$

Search keywords

factor of safety, reliability series parallel, FMEA, failure rate, design process

Top traps

- Adding reliabilities, or multiplying them for a parallel set.
- Swapping verification and validation.
- Thinking a high RPN is the only trigger (a severity of 9–10 is always critical).

1.D Physical Properties of Materials

1. Key concepts

- Tension test: the elastic slope is E, then the 0.2% offset gives yield Sy, the peak gives the ultimate Su, then necking and fracture. Ductility is measured by %elongation or %reduction of area.
- True stress $\sigma_T = \sigma(1+\varepsilon)$ and true strain $\varepsilon_T = \ln(1+\varepsilon)$ hold up to necking (constant volume). Strain hardening: $\sigma_T = K\varepsilon_T^n$, and necking starts at $\varepsilon_T = n$.
- Resilience (elastic energy per volume) $u_r = Sy^2/2E$. Toughness is the area under the whole curve, approximately $(Sy+Su)/2 \cdot \varepsilon_f$.
- Hardness tracks strength for steels: Su ≈ 0.5·HB ksi (3.45·HB MPa).
- Fe–C diagram: eutectoid 0.76% C at 727 °C (pearlite). Hypoeutectoid steels hold proeutectoid ferrite plus pearlite; use the lever rule. Quenching gives martensite and tempering restores toughness.
- BCC metals (ferritic steels) have a ductile-to-brittle transition. FCC metals (Al, Cu, austenitic stainless) do not. Creep matters above about 0.4 T_m (absolute); the Larson–Miller parameter is LMP = T(C + log t).
- Thermal expansion: $\delta = \alpha L \Delta T$. Galvanic corrosion attacks the anode (e.g. steel against copper); avoid a small anode with a large cathode.
- Composites: the rule of mixtures $E_L = E_f V_f + E_m(1-V_f)$ applies along the fibers (iso-strain). Across the fibers (iso-stress): $1/E_T = V_f/E_f + V_m/E_m$.

2. Essential formulas

Name	Formula	Variables / units
Hooke	$\sigma = E\varepsilon$; $\varepsilon = \delta/L$	E: psi or GPa
Poisson	$\varepsilon_{lat} = -\nu \cdot \varepsilon_{axial}$	ν ≈ 0.3 for steel
True stress/strain	$\sigma_T = \sigma(1+\varepsilon)$; $\varepsilon_T = \ln(1+\varepsilon)$	before necking
Reduction of area	$\%RA = (A_0 - A_f)/A_0 = 1 - (d_f/d_0)^2$	
Resilience	$u_r = Sy^2/(2E)$	in-lb/in³
Lever rule	$W_\alpha = (C_{eut} - C_0)/(C_{eut} - C_\alpha)$	wt% C
Larson–Miller	$LMP = T(C + \log t)$	T absolute (°R or K), t in h, C ≈ 20
Thermal strain	$\delta = \alpha L \Delta T$	α per °F or °C
Rule of mixtures	$E_L = E_f V_f + E_m(1-V_f)$	

3. Handbook search keywords

stress-strain diagram · true stress · modulus of resilience · hardness conversion · iron-carbon phase diagram · lever rule · Larson-Miller · coefficient of thermal expansion · galvanic series

4. Worked examples

Example 1: Tension test reduction
A 0.500 in diameter steel bar ($E = 30\times10^6$ psi, $\nu = 0.29$) carries 12,000 lb. Find the stress, the axial strain and the change in diameter.

1. $A = \pi(0.5)^2/4 = 0.1963$ in²; $\sigma = 12{,}000/0.1963 = 61{,}115$ psi.
2. $\varepsilon = \sigma/E = 2.037\times10^{-3}$.
3. $\Delta d = -\nu \cdot \varepsilon \cdot d = -0.29 \times 2.037\times10^{-3} \times 0.5 = -2.95\times10^{-4}$ in (it contracts).

Answer: σ ≈ 61,115 psi; Δd ≈ 295 µin decrease

Example 2: Lever rule
A 0.30% C steel is slowly cooled to just below 727 °C (α at 0.022% C, eutectoid at 0.76% C). Find the fraction

of proeutectoid ferrite.

1. $W_{\alpha'} = (0.76 - 0.30)/(0.76 - 0.022) = 0.623$.
2. Pearlite = 0.377.

Answer: 0.623

Example 3: Larson–Miller
LMP = 38,000 (°R, h, C = 20). Find the rupture life at 1,200 °F.

1. T = 1,200 + 460 = 1,660 °R.
2. $\log t = 38{,}000/1{,}660 - 20 = 2.892 \rightarrow t = 779$ h.

Answer: ≈ 779 h

5. Common traps

- Using °F or °C instead of absolute temperature in Larson–Miller.
- Using diameters instead of areas in %RA (square the diameter ratio).
- Using true stress after necking with $\sigma(1+\varepsilon)$ (the formula is invalid there).
- Putting the lever-rule numerator on the wrong side (use the OPPOSITE arm).
- Expecting FCC metals to show a DBTT.

1.D Physical Properties of Materials: one-page cheat summary

- ► $\sigma = P/A, \varepsilon = \delta/L, E = \sigma/\varepsilon$
- ► $\sigma_T = \sigma(1+\varepsilon), \varepsilon_T = \ln(1+\varepsilon)$
- ► $\%RA = 1 - (d_f/d_0)^2$
- ► $u_r = Sy^2/2E$
- ► $Su \approx 0.5\ HB\ ksi$
- ► Eutectoid 0.76% C, 727 °C
- ► $LMP = T_{abs}(20 + \log t)$
- ► $\delta = \alpha L \Delta T$
- ► $E_L = E_f V_f + E_m V_m$

Formulas

Hooke	$\sigma = E\varepsilon\ ;\ \varepsilon = \delta/L$
Poisson	$\varepsilon_{lat} = -\nu\varepsilon_{axial}$
True stress/strain	$\sigma_T = \sigma(1+\varepsilon);\ \varepsilon_T = \ln(1+\varepsilon)$
Reduction of area	$\%RA = (A_0 - A_f)/A_0 = 1 - (d_f/d_0)^2$
Resilience	$u_r = Sy^2/(2E)$
Lever rule	$W_\alpha = (C_{eut} - C_0)/(C_{eut} - C_\alpha)$
Larson–Miller	$LMP = T(C + \log t)$
Thermal strain	$\delta = \alpha L \Delta T$
Rule of mixtures	$E_L = E_f V_f + E_m(1-V_f)$

Search keywords

stress-strain diagram, true stress, modulus of resilience, hardness conversion, iron-carbon phase diagram, lever rule, Larson-Miller, coefficient of thermal expansion, galvanic series

Top traps

- Using °F or °C instead of absolute temperature in Larson–Miller.
- Using diameters instead of areas in %RA (square the diameter ratio).
- Using true stress after necking with σ(1+ε) (the formula is invalid there).

2.A Axial Loading

1. Key concepts

- Normal stress σ = P/A acts on a section perpendicular to the load. Elongation δ = PL/(AE) for each uniform segment; add the segments of a stepped bar.
- Statically indeterminate bars: write equilibrium, then compatibility (equal deformations, or deformations that sum to a gap), then the force–deformation relations.
- Parallel members sharing a rigid plate split the load in proportion to their axial stiffness k = AE/L.
- Poisson effect: lateral strain = −ν × axial strain. Diameter change Δd = −ν·ε·d.

2. Essential formulas

Name	Formula	Variables / units
Axial stress	$\sigma = P/A$	psi, MPa
Elongation	$\delta = PL/(AE)$; $\delta_total = \Sigma\, P_i L_i/(A_i E_i)$	in, mm
Axial stiffness	$k = AE/L$	
Load sharing (parallel)	$P_1 = P{\cdot}k_1/(k_1{+}k_2)$	equal strain
Poisson	$\Delta d = -\nu(\sigma/E)d$	

3. Handbook search keywords

axial deformation PL/AE · statically indeterminate · compatibility · Poisson's ratio

4. Worked examples

Example 1: Stepped bar
A steel bar (E = 30×10⁶ psi) has 12 in of Ø1.5 in and 8 in of Ø1.0 in and carries 20,000 lb tension. Find the total elongation.

1. A_1 = 1.7671 in², A_2 = 0.7854 in².
2. δ = (20,000/30×10⁶)(12/1.7671 + 8/0.7854) = 0.01132 in.

Answer: 0.0113 in

Example 2: Steel and aluminum posts
A rigid cap sits on a steel post (A = 1.0 in², E = 30×10⁶) and an aluminum post (A = 1.5 in², E = 10×10⁶) of equal length and carries 30,000 lb. Find the steel load.

1. Compatibility: equal shortening, so the load splits in proportion to AE.
2. P_s = 30,000·(30)/(30 + 15) = 20,000 lb; P_a = 10,000 lb.

Answer: 20,000 lb

5. Common traps

- Using one area for a stepped bar.
- Forgetting compatibility in indeterminate problems.
- Unit slips: kN with mm and GPa (1 GPa = 1,000 N/mm²).

2.A Axial Loading: one-page cheat summary

- ► $\sigma = P/A$
- ► $\delta = \Sigma PL/AE$
- ► $k = AE/L$
- ► Parallel: $P \propto AE/L$
- ► $\Delta d = -\nu(\sigma/E)d$

Formulas

Axial stress	$\sigma = P/A$
Elongation	$\delta = PL/(AE)$; $\delta_total = \Sigma\ P_iL_i/(A_iE_i)$
Axial stiffness	$k = AE/L$
Load sharing (parallel)	$P_1 = P{\cdot}k_1/(k_1+k_2)$
Poisson	$\Delta d = -\nu(\sigma/E)d$

Search keywords

axial deformation PL/AE, statically indeterminate, compatibility, Poisson's ratio

Top traps

- Using one area for a stepped bar.
- Forgetting compatibility in indeterminate problems.
- Unit slips: kN with mm and GPa (1 GPa = 1,000 N/mm²).

2.B Shear and Transverse Loading

1. Key concepts

- Direct shear (pins, bolts, keys): τ = V/A_shear. Single shear uses 1 area per pin; double shear uses 2.
- Beam transverse shear: τ = VQ/(Ib). It peaks at the neutral axis and is zero at the top and bottom fibers.
- Rectangle: τ_max = 1.5V/A. Solid circle: τ_max = (4/3)V/A. Thin-walled tube: τ_max ≈ 2V/A.
- Shear flow q = VQ/I (force per length) sizes the nails, bolts or welds that join built-up beam parts.

2. Essential formulas

Name	Formula	Variables / units
Direct shear	$\tau = V/(nA)$	n = number of shear planes
Transverse shear	$\tau = VQ/(Ib)$	Q = first moment of the area beyond the cut
Rectangle max	$\tau_{max} = 3V/(2A)$	
Circle max	$\tau_{max} = 4V/(3A)$	
Shear flow	$q = VQ/I$; fastener spacing $s = F_{allow}/q$	lb/in

3. Handbook search keywords

transverse shear stress VQ/Ib · shear flow · double shear · first moment of area

4. Worked examples

Example 1: Rectangular beam
A 3 in × 8 in rectangular beam carries V = 9,600 lb. Find τ_max.

1. A = 24 in².
2. τ_max = 1.5(9,600)/24 = 600 psi at the neutral axis.

Answer: 600 psi

Example 2: Clevis pin in double shear
A Ø0.625 in pin carries 8,000 lb in double shear. Find τ.

1. A = π(0.625)²/4 = 0.3068 in².
2. τ = 8,000/(2 × 0.3068) = 13,038 psi.

Answer: 13,038 psi

5. Common traps

- Using V/A (the average) when the maximum is asked.
- Forgetting double shear (2 planes).
- Using 4/3 for a rectangle or 3/2 for a circle.

2.B Shear and Transverse Loading: one-page cheat summary

► $\tau = V/(nA)$
► $\tau = VQ/Ib$
► Rect $1.5V/A$; circle $4V/3A$; tube $2V/A$
► $q = VQ/I$

Formulas

Direct shear	$\tau = V/(nA)$
Transverse shear	$\tau = VQ/(Ib)$
Rectangle max	$\tau_{max} = 3V/(2A)$
Circle max	$\tau_{max} = 4V/(3A)$
Shear flow	$q = VQ/I$; fastener spacing $s = F_{allow}/q$

Search keywords

transverse shear stress VQ/Ib, shear flow, double shear, first moment of area

Top traps

- Using V/A (the average) when the maximum is asked.
- Forgetting double shear (2 planes).
- Using 4/3 for a rectangle or 3/2 for a circle.

2.C Bending

1. Key concepts

- Draw shear and moment diagrams. Shear is the slope of the moment curve, so moment peaks where V = 0. Jumps happen at point loads (V) and applied couples (M).
- Flexure formula: σ = Mc/I = M/S, with tension and compression on opposite sides of the neutral axis.
- Parallel axis: I = I_c + Ad². For a composite section, find the centroid first.
- Deflection: use the handbook beam tables and superpose load cases. Simply supported with a center load: PL³/48EI. Cantilever with an end load: PL³/3EI. UDL on a simple span: 5wL⁴/384EI.

2. Essential formulas

Name	Formula	Variables / units
Flexure	$\sigma = Mc/I = M/S$	S = I/c
Rectangle	$I = bh^3/12$; $S = bh^2/6$	
Circle	$I = \pi d^4/64$; $S = \pi d^3/32$	
Parallel axis	$I = I_c + Ad^2$	
SS center load	$M_{max} = PL/4$; $\delta = PL^3/48EI$	
SS UDL	$M_{max} = wL^2/8$; $\delta = 5wL^4/384EI$	
Cantilever end load	$M_{max} = PL$; $\delta = PL^3/3EI$	
Cantilever UDL	$M_{max} = wL^2/2$; $\delta = wL^4/8EI$	

3. Handbook search keywords

flexure formula Mc/I · shear and moment diagrams · beam deflection tables · section modulus · parallel axis theorem

4. Worked examples

Example 1: Wood beam with UDL
A 12 ft simple span carries 400 lb/ft. The section is 3.5 in × 9.25 in. Find σ_max.

1. M = wL²/8 = 400(12)²/8 = 7,200 ft-lb = 86,400 in-lb.
2. S = bh²/6 = 49.9 in³.
3. σ = M/S = 1,731 psi.

Answer: 1,731 psi

Example 2: Midspan deflection
A simply supported steel beam (E = 29×10⁶ psi, I = 100 in⁴) spans 120 in with 10,000 lb at midspan. Find δ.

1. δ = PL³/48EI = 10,000(120)³/(48·29×10⁶·100) = 0.1241 in.

Answer: 0.1241 in

5. Common traps

- Leaving the moment in ft-lb while S is in in³ (multiply by 12).
- Using c = h instead of h/2.
- Taking the moment at the wrong location (max M is where V = 0).
- Mixing the cantilever and simply supported formulas.

2.C Bending: one-page cheat summary

- ► $\sigma = Mc/I = M/S$
- ► Rect $S = bh^2/6$
- ► Max M where V = 0
- ► SS: $PL/4, wL^2/8$
- ► δ: $PL^3/48EI, 5wL^4/384EI, PL^3/3EI, wL^4/8EI$
- ► $I = I_c + Ad^2$

Formulas

Flexure	$\sigma = Mc/I = M/S$
Rectangle	$I = bh^3/12$; $S = bh^2/6$
Circle	$I = \pi d^4/64$; $S = \pi d^3/32$
Parallel axis	$I = I_c + Ad^2$
SS center load	$M_{max} = PL/4$; $\delta = PL^3/48EI$
SS UDL	$M_{max} = wL^2/8$; $\delta = 5wL^4/384EI$
Cantilever end load	$M_{max} = PL$; $\delta = PL^3/3EI$
Cantilever UDL	$M_{max} = wL^2/2$; $\delta = wL^4/8EI$

Search keywords

flexure formula Mc/I, shear and moment diagrams, beam deflection tables, section modulus, parallel axis theorem

Top traps

- Leaving the moment in ft-lb while S is in in^3 (multiply by 12).
- Using c = h instead of h/2.
- Taking the moment at the wrong location (max M is where V = 0).

2.D Buckling

1. Key concepts

- Long slender columns fail by elastic buckling (Euler). Short columns fail by yielding. Intermediate columns follow the Johnson parabola.
- Effective length L_e = KL. Theoretical K: fixed–fixed 0.5, fixed–pinned 0.7, pinned–pinned 1.0, fixed–free 2.0. Design (recommended) K values are a little higher.
- Slenderness ratio = KL/r, with r = √(I/A). Compare it with the transition value C_c = √(2π²E/Sy). Above C_c use Euler; below it use Johnson.
- Always buckle about the weak axis (smallest I).

2. Essential formulas

Name	Formula	Variables / units
Euler load	$P_{cr} = \pi^2 EI/(KL)^2$	
Euler stress	$\sigma_{cr} = \pi^2 E/(KL/r)^2$	
Radius of gyration	$r = \sqrt{(I/A)}$; solid round $r = d/4$	
Transition	$C_c = \sqrt{(2\pi^2 E/Sy)}$	
Johnson	$P_{cr} = A[Sy - (Sy \cdot KL/r)^2/(4\pi^2 E)]$	intermediate

3. Handbook search keywords

Euler buckling · effective length factor · slenderness ratio · Johnson formula · radius of gyration

4. Worked examples

Example 1: Pinned steel column
A Ø1.25 in solid steel rod (E = 29×10⁶ psi, Sy = 50 ksi), 60 in long and pinned at both ends. Find P_cr.

1. r = d/4 = 0.3125 in; KL/r = 60/0.3125 = 192.
2. C_c = √(2π²·29×10⁶/50,000) = 107. Since 192 > 107, use Euler.
3. I = πd⁴/64 = 0.1198 in⁴; P_cr = π²EI/L² = 9,528 lb.

Answer: 9,528 lb

5. Common traps

- Using Euler for a short column (check KL/r against C_c).
- Using the strong-axis I.
- Swapping K for fixed–free (2.0) and fixed–fixed (0.5).

2.D Buckling: one-page cheat summary

- ► $P_{cr} = \pi^2 EI/(KL)^2$
- ► K: 0.5 / 0.7 / 1 / 2
- ► $r = \sqrt{(I/A)}$, round $d/4$
- ► $C_c = \sqrt{(2\pi^2 E/Sy)}$
- ► Johnson below C_c

Formulas

Euler load	$P_{cr} = \pi^2 EI/(KL)^2$
Euler stress	$\sigma_{cr} = \pi^2 E/(KL/r)^2$
Radius of gyration	$r = \sqrt{(I/A)}$; solid round $r = d/4$
Transition	$C_c = \sqrt{(2\pi^2 E/Sy)}$
Johnson	$P_{cr} = A[Sy - (Sy \cdot KL/r)^2/(4\pi^2 E)]$

Search keywords

Euler buckling, effective length factor, slenderness ratio, Johnson formula, radius of gyration

Top traps

- Using Euler for a short column (check KL/r against C_c).
- Using the strong-axis I.
- Swapping K for fixed–free (2.0) and fixed–fixed (0.5).

2.E Torsion

1. Key concepts

- Circular shafts: τ = Tc/J peaks at the outer surface. Angle of twist θ = TL/(JG) in radians.
- Power and torque: T(in-lb) = 63,025·hp/rpm; T(ft-lb) = 5,252·hp/rpm; T(N·m) = 9,549·kW/rpm.
- Hollow shafts: J = π(D⁴ − d⁴)/32. For the same weight they carry more torque.
- Thin-walled closed tube: τ = T/(2A_m t), where A_m is the area enclosed by the mid-wall line.

2. Essential formulas

Name	Formula	Variables / units
Torsion stress	$\tau = Tc/J = 16T/(\pi d^3)$ (solid)	
Polar moment	$J = \pi d^4/32$; hollow $\pi(D^4-d^4)/32$	
Twist	$\theta = TL/(JG)$ [rad]	G steel ≈ 11.5×10⁶ psi = 80 GPa
Power	$T = 63{,}025\ hp/n$ (in-lb)	
Thin tube	$\tau = T/(2A_m\ t)$	

3. Handbook search keywords

torsion shear stress Tc/J · angle of twist · polar moment of inertia · power torque speed

4. Worked examples

Example 1: Size a shaft for torque
A shaft transmits 20 hp at 600 rpm with an allowable τ of 12,000 psi. Find the minimum solid diameter.

1. T = 63,025(20)/600 = 2,101 in-lb.
2. d = (16T/πτ)^(1/3) = 0.962 in.

Answer: 0.962 in (use the next standard size)

Example 2: Twist
Same torque on a Ø1.25 in shaft, 48 in long, with G = 11.5×10⁶ psi. Find θ.

1. J = π(1.25)⁴/32 = 0.2397 in⁴.
2. θ = TL/JG = 0.0366 rad = 2.10°.

Answer: 2.10°

5. Common traps

- Leaving θ in radians when degrees are asked.
- Using I (πd⁴/64) instead of J.
- Using 5,252 (gives ft-lb) and then using in-lb.

2.E Torsion: one-page cheat summary

- ► $\tau = 16T/\pi d^3$
- ► $J = \pi d^4/32$
- ► $\theta = TL/JG$ (rad)
- ► $T = 63{,}025\ hp/n$ in-lb
- ► Thin tube $\tau = T/2A_m t$

Formulas

Torsion stress	$\tau = Tc/J = 16T/(\pi d^3)$ (solid)
Polar moment	$J = \pi d^4/32$; hollow $\pi(D^4 - d^4)/32$
Twist	$\theta = TL/(JG)$ [rad]
Power	$T = 63{,}025\ hp/n$ (in-lb)
Thin tube	$\tau = T/(2A_m t)$

Search keywords

torsion shear stress Tc/J, angle of twist, polar moment of inertia, power torque speed

Top traps

- Leaving θ in radians when degrees are asked.
- Using I ($\pi d^4/64$) instead of J.
- Using 5,252 (gives ft-lb) and then using in-lb.

2.F Combined Loading

1. Key concepts

- Superpose stresses at one point: axial P/A, bending ±Mc/I, torsion Tc/J, transverse VQ/Ib. Find the critical point (usually the outer fiber).
- Mohr's circle: center C = (σx+σy)/2, radius R = √[((σx−σy)/2)² + τ²]. $\sigma_{1,2}$ = C ± R and τ_max,in-plane = R.
- In 3-D, the absolute maximum shear = (σ_max − σ_min)/2, counting σ_3 = 0 in plane stress.
- Shaft with bending plus torsion: σ = 32M/πd³, τ = 16T/πd³, von Mises σ′ = √(σ² + 3τ²).

2. Essential formulas

Name	Formula	Variables / units
Principal stresses	$\sigma_{1,2} = (\sigma x+\sigma y)/2 \pm \sqrt{[((\sigma x-\sigma y)/2)^2 + \tau xy^2]}$	
Max in-plane shear	$\tau_max = \sqrt{[((\sigma x-\sigma y)/2)^2 + \tau xy^2]}$	
Principal angle	$\tan 2\theta p = 2\tau xy/(\sigma x-\sigma y)$	
Combined axial + bending	$\sigma = P/A \pm Mc/I$	
Shaft	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$	

3. Handbook search keywords

Mohr's circle · principal stresses · combined stresses · von Mises

4. Worked examples

Example 1: Principal stresses
σx = 12 ksi, σy = −4 ksi, τxy = 6 ksi. Find σ_1, σ_2 and τ_max.

1. C = 4 ksi; R = √(8² + 6²) = 10 ksi.
2. σ_1 = 14 ksi; σ_2 = -6 ksi; τ_max = 10 ksi.

Answer: σ_1 = 14, σ_2 = −6, τ_max = 10 ksi

5. Common traps

- Forgetting σ_3 = 0 when σ_1 and σ_2 have the same sign (then τ_abs = σ_1/2).
- Sign errors for compressive σy.
- Using √(σ² + 4τ²) (Tresca) when von Mises is asked.

2.F Combined Loading: one-page cheat summary

- ► $C = (\sigma x + \sigma y)/2$
- ► $R = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau^2]}$
- ► $\sigma_{1,2} = C \pm R$
- ► $\tau_max = R$ (in-plane)
- ► $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Formulas

Principal stresses	$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$
Max in-plane shear	$\tau_max = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$
Principal angle	$\tan 2\theta p = 2\tau xy/(\sigma x - \sigma y)$
Combined axial + bending	$\sigma = P/A \pm Mc/I$
Shaft	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Search keywords

Mohr's circle, principal stresses, combined stresses, von Mises

Top traps

- Forgetting $\sigma_3 = 0$ when σ_1 and σ_2 have the same sign (then $\tau_abs = \sigma_1/2$).
- Sign errors for compressive σy.
- Using $\sqrt{(\sigma^2 + 4\tau^2)}$ (Tresca) when von Mises is asked.

2.G Static Failure

1. Key concepts

- Ductile materials: use distortion energy (von Mises, the best fit) or maximum shear stress (Tresca, conservative). Brittle materials: use maximum normal stress, Coulomb–Mohr or modified Mohr.
- von Mises plane stress: $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)} = \sqrt{(\sigma x^2 - \sigma x \sigma y + \sigma y^2 + 3\tau^2)}$. n = Sy/σ′.
- Tresca: n = Sy/(σ_max − σ_min), including $\sigma_3 = 0$. Shear yield: von Mises Ssy = 0.577Sy; Tresca 0.5Sy.
- Static stress concentration: ductile materials can usually ignore Kt (local yielding relieves it). Brittle materials must apply Kt.

2. Essential formulas

Name	Formula	Variables / units
von Mises	$\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$; $n = Sy/\sigma'$	
Tresca	$n = Sy/(\sigma_1 - \sigma_3)$	
Shear yield	$Ssy = 0.577Sy\ (DE), 0.5Sy\ (MSS)$	
Max normal (brittle)	$n = Sut/\sigma_1$	

3. Handbook search keywords

distortion energy theory · maximum shear stress theory · factor of safety · Mohr theory brittle

4. Worked examples

Example 1: von Mises vs Tresca
Use σ_1 = 14 ksi and σ_2 = −6 ksi from the 2.F example, with Sy = 50 ksi. Find n by both theories.

1. $\sigma' = \sqrt{(14^2 - 14(-6) + 6^2)} = 17.78$ ksi → n = 2.81.
2. Tresca: $\sigma_1 - \sigma_3 = 14 - (-6) = 20$ ksi → n = 2.50.

Answer: n_DE = 2.81; n_Tresca = 2.50

5. Common traps

- Using Tresca with $\sigma_1 - \sigma_2$ when both are positive ($\sigma_3 = 0$ governs).
- Applying Kt to a ductile part under static load.
- Applying the brittle theories to steel.

2.G Static Failure: one-page cheat summary

► $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$
► Tresca $n = Sy/(\sigma max - \sigma min)$
► Ssy: $0.577Sy$ / $0.5Sy$
► Brittle: Coulomb–Mohr
► Ductile static: ignore Kt

Formulas

von Mises	$\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$; $n = Sy/\sigma'$
Tresca	$n = Sy/(\sigma_1 - \sigma_3)$
Shear yield	$Ssy = 0.577Sy$ (DE), $0.5Sy$ (MSS)
Max normal (brittle)	$n = Sut/\sigma_1$

Search keywords

distortion energy theory, maximum shear stress theory, factor of safety, Mohr theory brittle

Top traps

- Using Tresca with $\sigma_1 - \sigma_2$ when both are positive ($\sigma_3 = 0$ governs).
- Applying Kt to a ductile part under static load.
- Applying the brittle theories to steel.

2.H Fatigue Failure

1. Key concepts

- The endurance limit of steel: Se′ ≈ 0.5Sut (Sut ≤ 200 ksi), or 100 ksi above that. Correct it with the Marin factors Se = ka·kb·kc·kd·ke·Se′ (surface, size, load, temperature, reliability).
- Fluctuating stress: σa = (σmax − σmin)/2, σm = (σmax + σmin)/2. Apply K_f = 1 + q(Kt − 1) to the stresses.
- Mean-stress criteria: modified Goodman 1/n = σa/Se + σm/Sut (conservative, the usual exam choice). Gerber is a parabola (better fit). ASME-elliptic uses Sy. Also check first-cycle yield: σa + σm ≤ Sy/n.
- Finite life: S-N line through (10^3, f·Sut) and (10^6, Se), with S = aN^b. Miner's rule: $\Sigma n_i/N_i = 1$.

2. Essential formulas

Name	Formula	Variables / units
Endurance limit	$Se' = 0.5Sut\ (\text{steel})$	
Surface factor	$ka = a \cdot Sut^{b}\ (\text{machined: } a = 2.70,\ b = -0.265\ \text{ksi})$	
Size factor	$kb = 0.879d^{-0.107}\ (0.11\text{–}2\ \text{in})$	
Fatigue Kt	$K_f = 1 + q(Kt - 1)$	
Goodman	$1/n = \sigma a/Se + \sigma m/Sut$	
Basquin	$S_f = aN^{b}\ ;\ b = -(1/3)\log(f Sut/Se)\ ;\ a = (f Sut)^2/Se$	
Miner	$\Sigma\ n_i/N_i = 1$	

3. Handbook search keywords

endurance limit · Marin factors · modified Goodman · notch sensitivity · S-N diagram · Miner's rule

4. Worked examples

Example 1: Goodman factor of safety
A machined Ø1.0 in steel shaft (Sut = 90 ksi) sees σa = 18 ksi and σm = 20 ksi (K_f already applied), in rotating bending. Find n by Goodman.

1. ka = 2.70(90)^−0.265 = 0.819; kb = 0.879(1.0)^−0.107 = 0.879.
2. Se = 0.5(90)(0.819)(0.879) = 32.4 ksi.
3. 1/n = 18/32.4 + 20/90 → n = 1.29.

Answer: n ≈ 1.29

Example 2: Fatigue stress concentration
Kt = 2.4 and q = 0.85. Find K_f.

1. K_f = 1 + 0.85(1.4) = 2.19.

Answer: 2.19

5. Common traps

- Forgetting the Marin factors (using the raw 0.5Sut).
- Applying K_f to the strength instead of the stress (or applying it twice).
- Using σmax instead of σa and σm.
- Skipping the first-cycle yield check.

2.H Fatigue Failure: one-page cheat summary

- ▶ $Se' = 0.5Sut$
- ▶ $Se = ka \cdot kb \cdot kc \cdot kd \cdot ke \cdot Se'$
- ▶ $K_f = 1 + q(Kt - 1)$
- ▶ $\sigma a = (max - min)/2$; $\sigma m = (max + min)/2$
- ▶ Goodman: $1/n = \sigma a/Se + \sigma m/Sut$
- ▶ Miner: $\Sigma n/N = 1$

Formulas

Endurance limit	$Se' = 0.5Sut$ (steel)
Surface factor	$ka = a \cdot Sut^b$ (machined: a = 2.70, b = −0.265 ksi)
Size factor	$kb = 0.879d^{-0.107}$ (0.11–2 in)
Fatigue Kt	$K_f = 1 + q(Kt - 1)$
Goodman	$1/n = \sigma a/Se + \sigma m/Sut$
Basquin	$S_f = aN^b$; $b = -(1/3)\log(f \cdot Sut/Se)$; $a = (f \cdot Sut)^2/Se$
Miner	$\Sigma\, n_i/N_i = 1$

Search keywords

endurance limit, Marin factors, modified Goodman, notch sensitivity, S-N diagram, Miner's rule

Top traps

- Forgetting the Marin factors (using the raw 0.5Sut).
- Applying K_f to the strength instead of the stress (or applying it twice).
- Using σmax instead of σa and σm.

2.I Thermal and Interference Stresses

1. Key concepts

- A fully restrained bar heated by ΔT: σ = EαΔT (compressive). If there is a gap, only the expansion beyond the gap is restrained.
- Press and shrink fits: the interface pressure p comes from the interference δ (diametral). Same material, solid shaft: p = Eδ(D_o² − d²)/(2dD_o²).
- Hub hoop stress at the bore is σ_t = p(D_o² + d²)/(D_o² − d²). Torque capacity T = πμpd²L/2. Axial force capacity F = πμpdL.
- Shrink-fit assembly: the hub temperature rise is ΔT = (δ + clearance)/(αd).

2. Essential formulas

Name	Formula	Variables / units
Restrained bar	$\sigma = E\alpha\Delta T$	with gap: σ = E(αΔTL − gap)/L
Interference pressure	$p = E\delta(D_o^2 - d^2)/(2dD_o^2)$	same E, solid shaft
Hub bore hoop	$\sigma_t = p(D_o^2 + d^2)/(D_o^2 - d^2)$	
Torque capacity	$T = \pi\mu p d^2 L/2$	
Shrink temp	$\Delta T = (\delta + c)/(\alpha d)$	

3. Handbook search keywords

thermal stress · interference fit pressure · shrink fit · press fit torque

4. Worked examples

Example 1: Restrained rail
A steel bar (E = 29×10⁶ psi, α = 6.5×10⁻⁶/°F) is fixed between walls and heated 100 °F. Find σ.

1. σ = EαΔT = 29×10⁶(6.5×10⁻⁶)(100) = 18,850 psi (compression).

Answer: 18,850 psi

Example 2: Press-fit torque
A Ø2.0 in steel shaft is pressed into a steel hub (OD 4.0 in, L = 2.5 in) with δ = 0.002 in, E = 30×10⁶ psi, μ = 0.15. Find p and T.

1. p = 30×10⁶(0.002)(16 − 4)/(2·2·16) = 11,250 psi.
2. T = π(0.15)p(2²)(2.5)/2 = 26,507 in-lb.

Answer: T ≈ 26,507 in-lb

5. Common traps

- Using a radial interference where the formula expects diametral (or vice versa).
- Ignoring the gap.
- Forgetting that thermal stress does not depend on length when the bar is fully restrained.

2.I Thermal and Interference Stresses: one-page cheat summary

- ► $\sigma = E\alpha\Delta T$
- ► $p = E\delta(D_o^2 - d^2)/(2dD_o^2)$
- ► $T = \pi\mu p d^2 L/2$
- ► $\Delta T_{shrink} = (\delta + c)/(\alpha d)$

Formulas

Restrained bar	$\sigma = E\alpha\Delta T$
Interference pressure	$p = E\delta(D_o^2 - d^2)/(2dD_o^2)$
Hub bore hoop	$\sigma_t = p(D_o^2 + d^2)/(D_o^2 - d^2)$
Torque capacity	$T = \pi\mu p d^2 L/2$
Shrink temp	$\Delta T = (\delta + c)/(\alpha d)$

Search keywords

thermal stress, interference fit pressure, shrink fit, press fit torque

Top traps

- Using a radial interference where the formula expects diametral (or vice versa).
- Ignoring the gap.
- Forgetting that thermal stress does not depend on length when the bar is fully restrained.

3.A Bonds (Welding, Brazing, Adhesives)

1. Key concepts

- Fillet weld strength is set by the throat: t = 0.707h for an equal-leg fillet. Shear stress τ = F/(0.707hL).
- Weld groups under torsion or bending: treat the weld as a line. Find the unit properties J_u and I_u, then J = 0.707hJ_u. Add the primary (direct) and secondary (moment) shear as vectors.
- Weld symbols (AWS A2.4): a symbol below the reference line means arrow side; above means other side. A flag means field weld, a circle means all around, and the tail carries the process or spec.
- Adhesives are strong in shear and weak in peel. Lap joint: τ_avg = F/(bL).

2. Essential formulas

Name	Formula	Variables / units
Fillet throat stress	$\tau = F/(0.707hL)$	
Weld torsion	$\tau'' = Tr/J\ ;\ J = 0.707h \cdot J_u$	circle J_u = 2πr³
Weld bending	$\tau = Mc/I\ ;\ I = 0.707h \cdot I_u$	circle I_u = πr³
Adhesive lap	$\tau = F/(bL)$	

3. Handbook search keywords

fillet weld throat | weld group polar moment | welding symbols | adhesive lap joint

4. Worked examples

Example 1: Parallel fillet welds
Two 0.3125 in fillet welds, each 6 in long, carry 18,000 lb in shear. Find τ.

1. Throat area = 2(0.707)(0.3125)(6) = 2.651 in².
2. τ = 18,000/2.651 = 6,789 psi.

Answer: 6,789 psi

5. Common traps

- Using the leg h instead of the throat 0.707h.
- Adding the primary and secondary shear as scalars instead of vectors.
- Reversing arrow side and other side on the symbol.

3.A Bonds (Welding, Brazing, Adhesives): one-page cheat summary

- ► $t = 0.707h$
- ► $\tau = F/(0.707h\Sigma L)$
- ► $J = 0.707h \cdot J_u$
- ► Below line = arrow side

Formulas

Fillet throat stress	$\tau = F/(0.707hL)$
Weld torsion	$\tau'' = Tr/J$; $J = 0.707h \cdot J_u$
Weld bending	$\tau = Mc/I$; $I = 0.707h \cdot I_u$
Adhesive lap	$\tau = F/(bL)$

Search keywords

fillet weld throat, weld group polar moment, welding symbols, adhesive lap joint

Top traps

- Using the leg h instead of the throat 0.707h.
- Adding the primary and secondary shear as scalars instead of vectors.
- Reversing arrow side and other side on the symbol.

3.B Non-threaded Fasteners

1. Key concepts

- Pins, rivets and keys: check shear (single or double), bearing (crushing, σ_b = F/(dt)), and net-section tension of the plate, σ = F/((w − d)t).
- Keys: the force at the shaft surface is F = 2T/d. Shear τ = F/(wL); bearing on half the height σ = F/(L·h/2).
- Retaining rings, roll pins and shoulders locate parts axially.

2. Essential formulas

Name	Formula	Variables / units
Key force	$F = 2T/d$	
Key shear	$\tau = F/(wL)$	
Key bearing	$\sigma = 2F/(hL)$	square key h = w
Bearing on plate	$\sigma_b = F/(n \cdot d \cdot t)$	
Net section	$\sigma = F/((w - \Sigma d)t)$	

3. Handbook search keywords

key shear · rivet bearing stress · pin double shear · net section

4. Worked examples

Example 1: Square key length
A 0.375 in square key on a Ø1.5 in shaft carries 4,000 in-lb. Key Sy = 60 ksi, n = 2.5, Ssy = 0.577Sy. Find the minimum L.

1. F = 2(4,000)/1.5 = 5,333 lb.
2. Shear: L = Fn/(0.577·Sy·w) = 1.027 in.
3. Bearing: L = Fn/(Sy·w/2) = 1.185 in. Bearing governs.

Answer: 1.19 in

5. Common traps

- Using the diameter instead of the radius in F = T/r.
- Forgetting the bearing check on half the key height.

3.B Non-threaded Fasteners: one-page cheat summary

- ► $F = 2T/d$
- ► $\tau = F/wL$
- ► $\sigma_b = 2F/hL$
- ► $net = F/((w - d)t)$

Formulas

Key force	$F = 2T/d$
Key shear	$\tau = F/(wL)$
Key bearing	$\sigma = 2F/(hL)$
Bearing on plate	$\sigma_b = F/(n \cdot d \cdot t)$
Net section	$\sigma = F/((w - \Sigma d)t)$

Search keywords

key shear, rivet bearing stress, pin double shear, net section

Top traps

- Using the diameter instead of the radius in F = T/r.
- Forgetting the bearing check on half the key height.

3.C Threaded Fasteners

1. Key concepts

- Preload: F_i = 0.75A_tS_p for a reusable joint, 0.90A_tS_p for a permanent one. Torque T = KF_id with K ≈ 0.2 dry or 0.15 lubricated.
- Joint constant C = k_b/(k_b + k_m). The bolt picks up C·P of the external load and the members lose (1 − C)P. The joint separates at P_0 = F_i/(1 − C).
- Bolt fatigue: σ_a = CP/(2A_t) and σ_m = σ_a + F_i/A_t. Use Goodman with the preload line.
- Eccentric bolt groups: primary shear P/n plus secondary shear F = Mr/Σr², added as vectors.

2. Essential formulas

Name	Formula	Variables / units
Preload	$F_i = 0.75A_tS_p$ (reusable)	
Torque	$T = K \cdot F_i \cdot d$	
Joint constant	$C = k_b/(k_b + k_m)$	
Bolt load	$F_b = F_i + CP$	
Separation	$P_0 = F_i/(1 - C)$	
Eccentric group	$F'' = Mr_i/\Sigma r^2$	

3. Handbook search keywords

bolt preload · torque coefficient · joint stiffness constant · bolted joint separation · eccentric bolt group

4. Worked examples

Example 1: Preload, torque and separation
A 1/2-13 UNC grade 5 bolt (A_t = 0.1419 in², S_p = 85 ksi), reusable, K = 0.2, k_b = 3 and k_m = 10 (×10⁶ lb/in). Find F_i, T, the bolt load at P = 2,000 lb, and the separation load.

1. F_i = 0.75(0.1419)(85,000) = 9,046 lb.
2. T = 0.2(9,046)(0.5) = 905 in-lb.
3. C = 3/13 = 0.231; F_b = 9,046 + 0.231(2,000) = 9,508 lb.
4. P_0 = 9,046/(1 − 0.231) = 11,760 lb.

Answer: F_i ≈ 9,046 lb; T ≈ 905 in-lb; P_0 ≈ 11,760 lb

5. Common traps

- Giving the bolt the whole external load (it only takes C·P).
- Using the major-diameter area instead of A_t.
- Using 0.90 preload when the joint is reusable.

3.C Threaded Fasteners: one-page cheat summary

- ► $F_i = 0.75 A_t S_p$
- ► $T = K F_i d$
- ► $C = k_b/(k_b + k_m)$
- ► $F_b = F_i + CP$
- ► $P_0 = F_i/(1 - C)$

Formulas

Preload	$F_i = 0.75 A_t S_p$ (reusable)
Torque	$T = K \cdot F_i \cdot d$
Joint constant	$C = k_b/(k_b + k_m)$
Bolt load	$F_b = F_i + CP$
Separation	$P_0 = F_i/(1 - C)$
Eccentric group	$F'' = M r_i/\Sigma r^2$

Search keywords

bolt preload, torque coefficient, joint stiffness constant, bolted joint separation, eccentric bolt group

Top traps

- Giving the bolt the whole external load (it only takes $C \cdot P$).
- Using the major-diameter area instead of A_t.
- Using 0.90 preload when the joint is reusable.

4.A Gears and Gear Trains

1. Key concepts

- Geometry: diametral pitch P = N/d (teeth/in) or module m = d/N (mm). Center distance = $(d_1 + d_2)/2$. Mating gears must share P (or m) and the pressure angle.
- Forces (spur): $W_t = 33{,}000hp/V$ (V in ft/min), radial $W_r = W_t \cdot \tan\varphi$. Helical: axial $W_a = W_t \cdot \tan\psi$.
- Train value = product of driver teeth / product of driven teeth. Planetary: use the Willis formula $(n_L - n_A)/(n_F - n_A) = e$.
- Bending (Lewis): $\sigma = K_v W_t P/(FY)$. Wear (contact) uses the AGMA equations.

2. Essential formulas

Name	Formula	Variables / units
Pitch	$P = N/d\ ;\ m = d/N$	
Pitch-line velocity	$V = \pi dn/12\ (\text{ft/min})$	
Tangential load	$W_t = 33{,}000\ hp/V$	
Radial	$W_r = W_t \tan\varphi$	
Helical axial	$W_a = W_t \tan\psi$	
Lewis	$\sigma = K_v W_t P/(FY)$	
Planetary	$(n_{ring} - n_{arm})/(n_{sun} - n_{arm}) = -N_{sun}/N_{ring}$	

3. Handbook search keywords

diametral pitch · gear forces · Lewis bending · gear train value · planetary gear train

4. Worked examples

Example 1: Spur gear forces
An 18T pinion with P = 6 and φ = 20° transmits 5 hp at 1,150 rpm. Find W_t and W_r.

1. d = 18/6 = 3 in; V = π(3)(1,150)/12 = 903 ft/min.
2. $W_t = 33{,}000(5)/903 = 182.7$ lb; $W_r = 66.5$ lb.

Answer: $W_t \approx 183$ lb, $W_r \approx 66$ lb

5. Common traps

- Using V in ft/s.
- Mixing up driver and driven teeth in the train value.
- Using the pressure angle for the helical thrust (use the helix angle).

4.A Gears and Gear Trains: one-page cheat summary

- ► $P = N/d$; $C = (N_1 + N_2)/(2P)$
- ► $V = \pi dn/12$
- ► $W_t = 33{,}000hp/V$
- ► $W_r = W_t \tan\varphi$; $W_a = W_t \tan\psi$
- ► Planetary: Willis formula

Formulas

Pitch	$P = N/d$; $m = d/N$
Pitch-line velocity	$V = \pi dn/12$ (ft/min)
Tangential load	$W_t = 33{,}000\ hp/V$
Radial	$W_r = W_t \tan\varphi$
Helical axial	$W_a = W_t \tan\psi$
Lewis	$\sigma = K_v W_t P/(FY)$
Planetary	$(n_{ring} - n_{arm})/(n_{sun} - n_{arm}) = -N_{sun}/N_{ring}$

Search keywords

diametral pitch, gear forces, Lewis bending, gear train value, planetary gear train

Top traps

- Using V in ft/s.
- Mixing up driver and driven teeth in the train value.
- Using the pressure angle for the helical thrust (use the helix angle).

4.B Bearings

1. Key concepts

- Rolling bearings: $L_{10} = (C/P)^a \times 10^6$ revolutions, with a = 3 for ball and 10/3 for roller bearings. Hours = rev/(60n).
- Equivalent load P = XF_r + YF_a when F_a/F_r > e. Required rating C = P(L_D/L_R)^(1/a).
- Reliability adjustment (Weibull): for R ≠ 0.90, the life scales with [ln(1/R)/ln(1/0.9)]^(1/b).
- Journal bearings: Sommerfeld number S = (r/c)²μN/P; the Petroff torque estimates friction.

2. Essential formulas

Name	Formula	Variables / units
Rating life	$L_{10} = (C/P)^a \times 10^6$ rev	
Hours	$L_h = L_{10}/(60n)$	
Equivalent load	$P = XF_r + YF_a$	
Required C	$C = P(L_h 60n/10^6)^{(1/a)}$	
Sommerfeld	$S = (r/c)^2 \mu N/P$	

3. Handbook search keywords

bearing L10 life · dynamic load rating · equivalent radial load · Sommerfeld number

4. Worked examples

Example 1: Roller bearing life
Roller bearing with C = 15,000 N, P = 4,000 N, n = 3,600 rpm. Find L_{10} in hours.

1. L_{10} = (15,000/4,000)^(10/3) × 10⁶ rev = 81,928,982 rev.
2. Hours = rev/(60 × 3,600) = 379 h.

Answer: 379 h

5. Common traps

- Using a = 3 for roller bearings.
- Forgetting the 60 min/h.
- Using the radial load alone when thrust is present.

4.B Bearings: one-page cheat summary

- ► $L_{10} = (C/P)^a\ 10^6$ rev
- ► a = 3 ball, 10/3 roller
- ► $P = XF_r + YF_a$
- ► $h = rev/60n$

Formulas

Rating life	$L_{10} = (C/P)^a \times 10^6$ rev
Hours	$L_h = L_{10}/(60n)$
Equivalent load	$P = XF_r + YF_a$
Required C	$C = P(L_h 60n/10^6)^{(1/a)}$
Sommerfeld	$S = (r/c)^2 \mu N/P$

Search keywords

bearing L10 life, dynamic load rating, equivalent radial load, Sommerfeld number

Top traps

- Using a = 3 for roller bearings.
- Forgetting the 60 min/h.
- Using the radial load alone when thrust is present.

4.C Belts, Chains, Clutches, Brakes, Power Screws

1. Key concepts

- Flat belt: $T_1/T_2 = e^{(\mu\theta)}$ at impending slip (V-belts use $\mu/\sin(\beta/2)$). Power = $(T_1 - T_2)V$.
- Chain: V = NpN_rpm/12 (ft/min). Chordal action is reduced by using more sprocket teeth.
- Disk clutch (uniform wear): T = NμF(D + d)/4. Uniform pressure gives slightly more torque. Band brake: T = $(T_1 - T_2)r$ with $T_1/T_2 = e^{(\mu\theta)}$.
- Square-thread power screw: T_raise = (Wd_m/2)(l + πμd_m)/(πd_m − μl) + collar torque Wμ_cd_c/2. It is self-locking if πμd_m > l. Efficiency = Wl/(2πT).

2. Essential formulas

Name	Formula	Variables / units
Belt	$T_1/T_2 = e^{(\mu\theta)}$; $hp = (T_1 - T_2)V/33{,}000$	
Open-belt length	$L = 2C + \pi(D + d)/2 + (D - d)^2/(4C)$	
Clutch (uniform wear)	$T = N\mu F(D + d)/4$	
Screw raise	$T = (Wd_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l)$	
Efficiency	$e = Wl/(2\pi T)$	

3. Handbook search keywords

belt tension ratio · band brake · disk clutch uniform wear · power screw torque · chain velocity

4. Worked examples

Example 1: Flat belt tensions
A belt transmits 5 hp from a 6 in pulley at 1,150 rpm, θ = 180°, μ = 0.3. Find T_1.

1. Net force $T_1 - T_2 = T/r = (63{,}025 \cdot 5/1{,}150)/3 = 91.3$ lb.
2. $e^{(\mu\theta)} = e^{(0.3\pi)} = 2.566$; $T_1 = F \cdot e/(e - 1) = 149.7$ lb.

Answer: 150 lb

5. Common traps

- Using θ in degrees in $e^{(\mu\theta)}$.
- Forgetting the collar friction on a power screw.
- Using uniform pressure when the question says uniform wear.

4.C Belts, Chains, Clutches, Brakes, Power Screws: one-page cheat summary

- ► $T_1/T_2 = e^{(\mu\theta)}$, θ in rad
- ► $hp = (T_1 - T_2)V/33{,}000$
- ► Clutch: $N\mu F(D + d)/4$
- ► Screw: self-locking if $\pi\mu d_m > l$

Formulas

Belt	$T_1/T_2 = e^{(\mu\theta)}$; $hp = (T_1 - T_2)V/33{,}000$
Open-belt length	$L = 2C + \pi(D + d)/2 + (D - d)^2/(4C)$
Clutch (uniform wear)	$T = N\mu F(D + d)/4$
Screw raise	$T = (Wd_m/2)(l + \pi\mu d_m)/(\pi d_m - \mu l)$
Efficiency	$e = Wl/(2\pi T)$

Search keywords

belt tension ratio, band brake, disk clutch uniform wear, power screw torque, chain velocity

Top traps

- Using θ in degrees in e^(μθ).
- Forgetting the collar friction on a power screw.
- Using uniform pressure when the question says uniform wear.

4.D Shafts and Keys

1. Key concepts

- Design steps: loads → shear and moment diagrams → critical sections (shoulders, keyways) → fatigue size → check deflection, slopes and critical speed.
- DE-Goodman with rotating bending (M_a) and steady torque (T_m): $d = \{16n/\pi\,[2K_{fM_a}/S_e + \sqrt{3}K_{fsT_m}/S_{ut}]\}^{(1/3)}$.
- Critical speed: $\omega = \sqrt{(g/\delta_{st})}$ for a single disk (Rayleigh for several). Keep the operating speed well away from it.
- Flywheel: $\Delta E = C_s I\omega^2$, with $C_s = (\omega_{max} - \omega_{min})/\omega_{avg}$.

2. Essential formulas

Name	Formula	Variables / units
DE-Goodman diameter	$d = \{16n/\pi\,[2K_{fM_a}/S_e + \sqrt{3}K_{fsT_m}/S_{ut}]\}^{(1/3)}$	
Static torsion	$d = (16Tn/(\pi S_{sy}))^{(1/3)}$	
Critical speed	$n_c = (30/\pi)\sqrt{(g/\delta)}$	$g = 386$ in/s²
Flywheel	$\Delta E = C_s I\omega^2$	

3. Handbook search keywords

shaft design DE-Goodman · critical speed · flywheel coefficient of fluctuation · key

4. Worked examples

Example 1: Critical speed
A 250 lb disk sits at the center of a Ø1.5 in steel shaft ($E = 30\times10^6$ psi) on bearings 48 in apart. Find n_c.

1. $I = \pi(1.5)^4/64 = 0.2485$ in⁴; $\delta = WL^3/48EI = 0.0773$ in.
2. $n_c = (30/\pi)\sqrt{(386.1/\delta)} = 675$ rpm.

Answer: 675 rpm

5. Common traps

- Using $g = 32.2$ ft/s² with δ in inches.
- Leaving out K_f at shoulders.
- Applying the mean-torsion term to the alternating stress.

4.D Shafts and Keys: one-page cheat summary

- ► DE-Goodman d formula
- ► $n_c = (30/\pi)\sqrt{(g/\delta)}$
- ► $\Delta E = C_s I \omega^2$
- ► Check slopes at bearings

Formulas

DE-Goodman diameter	$d = \{16n/\pi\ [2K_f M_a/S_e + \sqrt{3}K_{fs}T_m/S_{ut}]\}^{(1/3)}$
Static torsion	$d = (16Tn/(\pi S_{sy}))^{(1/3)}$
Critical speed	$n_c = (30/\pi)\sqrt{(g/\delta)}$
Flywheel	$\Delta E = C_s I \omega^2$

Search keywords

shaft design DE-Goodman, critical speed, flywheel coefficient of fluctuation, key

Top traps

- Using g = 32.2 ft/s² with δ in inches.
- Leaving out K_f at shoulders.
- Applying the mean-torsion term to the alternating stress.

4.E Motors and Engines

1. Key concepts

- Induction motor: synchronous speed n_s = 120f/p (p = poles). Slip s = (n_s − n)/n_s.
- Three-phase input: P = √3·V·I·pf. Output = η × input. T = 63,025hp/n.
- Engines: brake power = 2πNT. Thermal efficiency = output/fuel energy. BSFC = fuel rate/brake power.

2. Essential formulas

Name	Formula	Variables / units
Sync speed	$n_s = 120f/p$	
Slip	$n = n_s(1 - s)$	
3-phase power	$P = \sqrt{3} \cdot V \cdot I \cdot pf$	
Line current	$I = hp \cdot 746/(\sqrt{3} \cdot V \cdot \eta \cdot pf)$	

3. Handbook search keywords

induction motor slip · synchronous speed · three phase power · motor efficiency

4. Worked examples

Example 1: Motor current
A 25 hp, 460 V three-phase motor has η = 0.90 and pf = 0.85. Find the line current.

1. I = 25(746)/(√3·460·0.90·0.85) = 30.6 A.

Answer: 30.6 A

5. Common traps

- Using poles/2 incorrectly (n_s = 120f/p with p = total poles).
- Forgetting efficiency when going from output to input power.

4.E Motors and Engines: one-page cheat summary

- ► $n_s = 120f/p$
- ► $n = n_s(1 - s)$
- ► $P = \sqrt{3}VI \cdot pf$
- ► $T = 63{,}025hp/n$

Formulas

Sync speed	$n_s = 120f/p$
Slip	$n = n_s(1 - s)$
3-phase power	$P = \sqrt{3} \cdot V \cdot I \cdot pf$
Line current	$I = hp \cdot 746/(\sqrt{3} \cdot V \cdot \eta \cdot pf)$

Search keywords

induction motor slip, synchronous speed, three phase power, motor efficiency

Top traps

- Using poles/2 incorrectly ($n_s = 120f/p$ with p = total poles).
- Forgetting efficiency when going from output to input power.

5.A Pressurized Vessels and Piping

1. Key concepts

- Thin wall (t < D/20): hoop σ_t = pD/(2t), longitudinal σ_L = pD/(4t). Sphere: σ = pD/(4t).
- Thick wall (Lamé): σ_t = [p_ir_i² − p_or_o² + r_i²r_o²(p_i − p_o)/r²]/(r_o² − r_i²). Internal pressure only: the maximum is at r_i, σ_t = p(r_o² + r_i²)/(r_o² − r_i²).
- ASME VIII Div. 1 shell: t = PR/(SE − 0.6P), with R the inside radius and E the joint efficiency.

2. Essential formulas

Name	Formula	Variables / units
Hoop	$\sigma_t = pD/(2t)$	
Longitudinal	$\sigma_L = pD/(4t)$	
Lamé (internal)	$\sigma_{t,max} = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$	
ASME shell	$t = PR/(SE - 0.6P)$	

3. Handbook search keywords

thin-walled pressure vessel | Lamé thick-walled cylinder | ASME Section VIII thickness

4. Worked examples

Example 1: ASME shell thickness
R = 30 in, P = 250 psi, S = 20,000 psi, E = 0.85. Find t.

1. t = 250(30)/(20,000·0.85 − 0.6·250) = 0.445 in (add the corrosion allowance).

Answer: 0.445 in

5. Common traps

- Using the radius in pD/2t (D is the diameter).
- Using the thin-wall formula when t/r > 0.1.

5.A Pressurized Vessels and Piping: one-page cheat summary

► $\sigma_t = pD/2t$; $\sigma_L = pD/4t$

► Lamé max at the bore

► $t = PR/(SE - 0.6P)$

Formulas

Hoop	$\sigma_t = pD/(2t)$
Longitudinal	$\sigma_L = pD/(4t)$
Lamé (internal)	$\sigma_{t,max} = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$
ASME shell	$t = PR/(SE - 0.6P)$

Search keywords

thin-walled pressure vessel, Lamé thick-walled cylinder, ASME Section VIII thickness

Top traps

- Using the radius in pD/2t (D is the diameter).
- Using the thin-wall formula when t/r > 0.1.

5.B Hydraulic and Pneumatic Components

1. Key concepts

- Cylinder force F = pA. On retraction use the annulus A = π(D² − d²)/4. Speed v = Q/A.
- Hydraulic power: hp = Q(gpm)·p(psi)/1,714. Input = output/η.
- Pneumatics: the air is compressible, so use absolute pressure in gas-law calculations.

2. Essential formulas

Name	Formula	Variables / units
Force	$F = pA$	
Speed	$v = Q/A\ (231\ \text{in}^3/\text{gal})$	
Power	$hp = Q{\cdot}p/1{,}714$	

3. Handbook search keywords

hydraulic cylinder force · hydraulic power 1714 · pump efficiency

4. Worked examples

Example 1: Cylinder speed and force
A Ø3 in bore cylinder at 1,500 psi is supplied with 12 gpm. Find the extension force and speed.

1. A = 7.069 in²; F = 10,603 lb.
2. v = 12(231)/60/7.069 = 6.54 in/s.

Answer: F ≈ 10,603 lb; v ≈ 6.54 in/s

5. Common traps

- Using the full bore area on retraction.
- Gauge pressure in gas laws.

5.B Hydraulic and Pneumatic Components: one-page cheat summary

- ► $F = pA$
- ► $v = Q/A$
- ► $hp = Qp/1{,}714$

Formulas

Force	$F = pA$
Speed	$v = Q/A\ (231\ in^3/gal)$
Power	$hp = Q{\cdot}p/1{,}714$

Search keywords

hydraulic cylinder force, hydraulic power 1714, pump efficiency

Top traps

- Using the full bore area on retraction.
- Gauge pressure in gas laws.

5.C Beams, Trusses, and Frames (Statics)

1. Key concepts

- Free-body diagram first: isolate the body, show every external force and moment (weight, supports, friction). Unknown reactions: pin = 2 forces, roller = 1, fixed = 2 forces + 1 moment.
- Equilibrium: ΣF_x = 0, ΣF_y = 0, ΣM = 0. Take moments about the point with the most unknowns.
- Trusses: method of joints (ΣF = 0 at each pin; start where there are ≤ 2 unknowns) or method of sections (cut ≤ 3 members, then take moments). Zero-force members: at an unloaded joint where two members are collinear, the third member is zero.
- Friction: F ≤ μ_sN. Impending motion means F = μ_sN. Belt/capstan: $T_1/T_2 = e^{(\mu\theta)}$.
- Centroids and moments of inertia of composites: $\bar{x} = \Sigma A\, x/\Sigma A$, then $I = \Sigma(I_c + Ad^2)$.

2. Essential formulas

Name	Formula	Variables / units
Equilibrium	$\Sigma F = 0\ ;\ \Sigma M = 0$	
Friction	$F = \mu N$ (impending)	
Capstan	$T_1 = T_2 e^{(\mu\theta)}$	θ rad
Centroid	$\bar{y} = \Sigma A_i y_i / \Sigma A_i$	
Parallel axis	$I = \Sigma(I_c + Ad^2)$	
Two-force member	force along the line joining its pins	

3. Handbook search keywords

equilibrium · method of joints · method of sections · zero-force member · friction · centroid · parallel axis

4. Worked examples

Example 1: Beam reaction
A 12 ft simple beam carries 9,000 lb at 4 ft from support A. Find R_A and R_B.

1. ΣM_B = 0: R_A(12) = 9,000(8).
2. R_A = 6,000 lb; R_B = 9,000 − 6,000 = 3,000 lb.

Answer: R_A = 6,000 lb, R_B = 3,000 lb

Example 2: Ladder friction
A uniform ladder leans at 60° from the floor against a smooth wall. Find the minimum μ at the floor.

1. Smooth wall: horizontal reaction N_w only. Floor: N = W, F = N_w.
2. ΣM about the foot: N_w·L sin60 = W(L/2)cos60 → N_w = W/(2 tan60).
3. μ_min = N_w/N = 1/(2 tan60°) = 0.289.

Answer: 0.289

Example 3: Truss member
Two members meet at an apex at 40° to the horizontal and carry 10,000 lb vertically. Find the force in each.

1. 2F sin40° = 10,000 → F = 7,779 lb (compression).

Answer: 7,779 lb

5. Common traps

- Leaving the friction force or the weight off the FBD.
- Adding a 'ma' force to a static FBD.
- Using cos where sin belongs (draw the angle).
- Taking moments about a point where the unknowns do not drop out.

5.C Beams, Trusses, and Frames (Statics): one-page cheat summary

► FBD → $\Sigma F = 0, \Sigma M = 0$
► Pin 2, roller 1, fixed 3 unknowns
► Zero-force: 2 collinear + 1 at an unloaded joint
► $F = \mu N$ impending
► $T_1/T_2 = e^{\wedge}(\mu\theta)$
► $\bar{y} = \Sigma Ay/\Sigma A$; $I = \Sigma(I_c + Ad^2)$

Formulas

Equilibrium	$\Sigma F = 0$; $\Sigma M = 0$
Friction	$F = \mu N$ (impending)
Capstan	$T_1 = T_2 e^{\wedge}(\mu\theta)$
Centroid	$\bar{y} = \Sigma A_i y_i / \Sigma A_i$
Parallel axis	$I = \Sigma(I_c + Ad^2)$
Two-force member	force along the line joining its pins

Search keywords

equilibrium, method of joints, method of sections, zero-force member, friction, centroid, parallel axis

Top traps

- Leaving the friction force or the weight off the FBD.
- Adding a 'ma' force to a static FBD.
- Using cos where sin belongs (draw the angle).

5.D Springs

1. Key concepts

- Helical compression spring: $k = Gd^4/(8D^3N_a)$. Spring index $C = D/d$ (4–12 is good).
- Stress: $\tau = K_W \cdot 8FD/(\pi d^3)$, with Wahl $K_W = (4C - 1)/(4C - 4) + 0.615/C$ (or Bergsträsser $K_B = (4C + 2)/(4C - 3)$).
- Series springs: $1/k = \Sigma 1/k_i$. Parallel: $k = \Sigma k_i$. Energy $U = \tfrac{1}{2}kx^2$.
- Spring fatigue: Zimmerli data with torsional Goodman. Surge frequency: avoid resonance with the operating frequency.

2. Essential formulas

Name	Formula	Variables / units
Rate	$k = Gd^4/(8D^3N)$	
Stress	$\tau = K \cdot 8FD/(\pi d^3)$	
Wahl	$K = (4C - 1)/(4C - 4) + 0.615/C$	
Series/parallel	$1/k = \Sigma 1/k_i$; $k = \Sigma k_i$	

3. Handbook search keywords

helical spring rate · Wahl factor · spring index · springs in series

4. Worked examples

Example 1: Spring rate
Music wire $G = 11.5\times10^6$ psi, $d = 0.148$ in, $D = 1.5$ in, $N_a = 8$. Find k.

1. $k = 11.5\times10^6(0.148)^4/(8\cdot1.5^3\cdot8) = 25.5$ lb/in.

Answer: 25.5 lb/in

5. Common traps

- Using the outside diameter for D (use the mean coil diameter).
- Leaving out the curvature factor.

5.D Springs: one-page cheat summary

- $k = Gd^4/8D^3N$
- $\tau = K8FD/\pi d^3$
- Series $1/k = \Sigma 1/k$

Formulas

Rate	$k = Gd^4/(8D^3N)$
Stress	$\tau = K8FD/(\pi d^3)$
Wahl	$K = (4C - 1)/(4C - 4) + 0.615/C$
Series/parallel	$1/k = \Sigma 1/k_i$; $k = \Sigma k_i$

Search keywords

helical spring rate, Wahl factor, spring index, springs in series

Top traps

- Using the outside diameter for D (use the mean coil diameter).
- Leaving out the curvature factor.

5.E Vibrating Systems

1. Key concepts

- SDOF: $\omega_n = \sqrt{(k/m)}$, $f_n = \omega_n/2\pi$. Static deflection shortcut: $\omega_n = \sqrt{(g/\delta_{st})}$, $f_n \approx 3.13/\sqrt{\delta}$(in) Hz.
- Damping: $c_c = 2\sqrt{(km)} = 2m\omega_n$, $\zeta = c/c_c$. Damped frequency $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$. Log decrement $\delta = (1/n)\ln(x_0/x_n)$, $\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}$.
- Forced response: $X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, with $r = \omega/\omega_n$. Resonance is near $r = 1$.
- Transmissibility $TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$. Isolation ($TR < 1$) needs $r > \sqrt{2}$. Above $\sqrt{2}$, damping INCREASES TR, and softer springs or more mass help. Undamped: $TR = 1/(r^2 - 1)$.
- Rotating unbalance: $X = (m_e e/M)r^2/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$.
- Equivalent systems: springs in series or parallel. Cantilever $k = 3EI/L^3$; simply supported center $k = 48EI/L^3$. Torsional $k_t = GJ/L$, $\omega_n = \sqrt{(k_t/J_{mass})}$. Compound pendulum $\omega_n = \sqrt{(mgd/I_O)}$.

2. Essential formulas

Name	Formula	Variables / units
Natural frequency	$\omega_n = \sqrt{(k/m)}\ ;\ f = \omega_n/2\pi$	
Static deflection	$\omega_n = \sqrt{(g/\delta_{st})}$	
Damping ratio	$\zeta = c/(2\sqrt{(km)})$	
Damped	$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$	
Log decrement	$\delta = (1/n)\ln(x_0/x_n)\ ;\ \zeta \approx \delta/2\pi$	
Amplitude	$X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$	
Transmissibility	$TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$	
Beam stiffness	cantilever $3EI/L^3$; SS center $48EI/L^3$	

3. Handbook search keywords

natural frequency · damping ratio · logarithmic decrement · transmissibility · vibration isolation · rotating unbalance · forced vibration

4. Worked examples

Example 1: Damping and frequency
m = 10 kg, k = 100 kN/m, c = 150 N·s/m. Find f_n, ζ and f_d.

1. $\omega_n = \sqrt{(100{,}000/10)} = 100.0$ rad/s → $f_n = 15.92$ Hz.
2. $c_c = 2\sqrt{(km)} = 2{,}000$ N·s/m; $\zeta = 150/2{,}000 = 0.075$.
3. $f_d = f_n\sqrt{(1 - \zeta^2)} = 15.87$ Hz (practically the same).

Answer: $f_n = 15.92$ Hz, $\zeta = 0.075$

Example 2: Isolation
A machine on mounts runs at $r = 3$ with $\zeta = 0.1$. Find TR. Would more damping help?

1. $TR = \sqrt{[(1 + 0.6^2)/((1 - 9)^2 + 0.6^2)]} = 0.1454$.
2. No. Above $r = \sqrt{2}$, extra damping raises TR.

Answer: $TR \approx 0.145$

Example 3: Isolator static deflection
Undamped mounts must give TR = 0.20 at 1,800 rpm. Find the required static deflection.

1. $r^2 = 1/TR + 1 = 6 \rightarrow r = 2.449$.
2. $\omega = 1{,}800 \cdot 2\pi/60 = 188.5$ rad/s; $\omega_n = \omega/r = 77.0$ rad/s.
3. $\delta = g/\omega_n^2 = 386.1/5922 = 0.0652$ in.

Answer: 0.065 in

5. Common traps

- Mixing Hz and rad/s ($\omega = 2\pi f$).
- Thinking damping always helps isolation (it hurts above $\sqrt{2}$).
- Leaving k in kN/m or N/mm while m is in kg.
- Using the series formula for parallel springs.

5.E Vibrating Systems: one-page cheat summary

- ► $\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta)}$
- ► $\zeta = c/2\sqrt{(km)}$
- ► $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$
- ► $\zeta \approx \ln(x_0/x_n)/(2\pi n)$
- ► TR < 1 only for $r > \sqrt{2}$
- ► Undamped $TR = 1/(r^2 - 1)$
- ► k: cantilever $3EI/L^3$, SS $48EI/L^3$

Formulas

Natural frequency	$\omega_n = \sqrt{(k/m)}$; $f = \omega_n/2\pi$
Static deflection	$\omega_n = \sqrt{(g/\delta_{st})}$
Damping ratio	$\zeta = c/(2\sqrt{(km)})$
Damped	$\omega_d = \omega_n\sqrt{(1-\zeta^2)}$
Log decrement	$\delta = (1/n)\ln(x_0/x_n)$; $\zeta \approx \delta/2\pi$
Amplitude	$X = (F_0/k)/\sqrt{[(1-r^2)^2 + (2\zeta r)^2]}$
Transmissibility	$TR = \sqrt{[(1+(2\zeta r)^2)/((1-r^2)^2 + (2\zeta r)^2)]}$
Beam stiffness	cantilever $3EI/L^3$; SS center $48EI/L^3$

Search keywords

natural frequency, damping ratio, logarithmic decrement, transmissibility, vibration isolation, rotating unbalance, forced vibration

Top traps

- Mixing Hz and rad/s ($\omega = 2\pi f$).
- Thinking damping always helps isolation (it hurts above $\sqrt{2}$).
- Leaving k in kN/m or N/mm while m is in kg.

5.F Basic Machines and Mechanisms (Dynamics)

1. Key concepts

- Kinematics: constant acceleration gives $v = v_0 + at$, $s = v_0t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2as$. Rotation is the same with θ, ω, α, and $v = r\omega$, $a_t = r\alpha$, $a_n = r\omega^2$.
- Kinetics: $\Sigma F = ma$ and $\Sigma M_G = I_G\alpha$. Draw the FBD and the kinetic diagram.
- Work–energy: $T_1 + \Sigma U = T_2$, with $T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. Spring energy $\frac{1}{2}kx^2$, gravity mgh, friction work μNd.
- Impulse–momentum: $mv_1 + \int F\,dt = mv_2$. Impact: momentum is conserved; $e = (v_2' - v_1')/(v_1 - v_2)$.
- Rolling without slip: $a = g\sin\theta/(1 + I/(mr^2))$. Hoop: $I = mr^2$. Solid disk: $\frac{1}{2}mr^2$. Sphere: $0.4mr^2$.
- Mechanisms: mechanical advantage, linkages (Grashof), cams, and gear/belt speed ratios.

2. Essential formulas

Name	Formula	Variables / units
Kinematics	$v^2 = v_0^2 + 2as\ ;\ s = v_0t + \frac{1}{2}at^2$	
Newton	$\Sigma F = ma\ ;\ \Sigma M = I\alpha$	
Work–energy	$\frac{1}{2}mv_1^2 + \Sigma U = \frac{1}{2}mv_2^2$	
Restitution	$e = (v_2' - v_1')/(v_1 - v_2)$	
Rolling	$a = g\sin\theta/(1 + k^2/r^2)$	
Centripetal	$F = mr\omega^2 = mv^2/r$	
Braking distance	$d = v^2/(2\mu g)$	

3. Handbook search keywords

kinematics · Newton's second law · work-energy · impulse momentum · coefficient of restitution · rolling without slipping · mass moment of inertia

4. Worked examples

Example 1: Crate on an incline
A crate is released on a 25° incline with $\mu_k = 0.15$. Find a and the speed after 4 m.

1. $a = g(\sin 25° - 0.15\cos 25°) = 2.81$ m/s².
2. $v = \sqrt{(2as)} = \sqrt{(2 \cdot 2.81 \cdot 4)} = 4.74$ m/s.

Answer: a = 2.81 m/s², v = 4.74 m/s

Example 2: Impact
A 3 kg ball at 12 m/s hits a stationary 8 kg ball, e = 0.8. Find v_2'.

1. Momentum: $36 = 3v_1' + 8v_2'$. Restitution: $v_2' - v_1' = 9.6$.
2. $v_2' = m_1v_1(1 + e)/(m_1 + m_2) = 3(12)(1.8)/11 = 5.89$ m/s.

Answer: 5.89 m/s

Example 3: Spinning up a rotor
I = 4 kg·m², net torque 50 N·m. Time to reach 1,200 rpm?

1. $\omega = 1{,}200 \cdot 2\pi/60 = 125.7$ rad/s; $\alpha = T/I = 12.5$ rad/s².
2. $t = \omega/\alpha = 10.05$ s.

Answer: 10.1 s

5. Common traps

- Forgetting the rotational KE of rolling bodies.
- Treating rpm as rad/s.
- Assuming energy is conserved in an impact (only momentum is).
- Getting the friction direction wrong (it opposes relative motion).

5.F Basic Machines and Mechanisms (Dynamics): one-page cheat summary

- ► $v^2 = v_0^2 + 2as$
- ► $\Sigma F = ma$, $\Sigma M = I\alpha$
- ► $T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$
- ► e = sep/approach
- ► Rolling $a = g\sin\theta/(1 + I/mr^2)$
- ► $\omega = 2\pi n/60$

Formulas

Kinematics	$v^2 = v_0^2 + 2as$; $s = v_0t + \frac{1}{2}at^2$
Newton	$\Sigma F = ma$; $\Sigma M = I\alpha$
Work–energy	$\frac{1}{2}mv_1^2 + \Sigma U = \frac{1}{2}mv_2^2$
Restitution	$e = (v_2' - v_1')/(v_1 - v_2)$
Rolling	$a = g\sin\theta/(1 + k^2/r^2)$
Centripetal	$F = mr\omega^2 = mv^2/r$
Braking distance	$d = v^2/(2\mu g)$

Search keywords

kinematics, Newton's second law, work-energy, impulse momentum, coefficient of restitution, rolling without slipping, mass moment of inertia

Top traps

- Forgetting the rotational KE of rolling bodies.
- Treating rpm as rad/s.
- Assuming energy is conserved in an impact (only momentum is).

6.A Manufacturing Methods

1. Key concepts

- Machining: spindle speed N = 12V/(πD) (V in sfm, D in in). Feed rate = f·N. Time = L/(fN). Taylor tool life $VT^n = C$.
- Process choice: casting (complex shapes, lower strength), forging (good grain flow, fatigue strength), machining (precision), additive (complex shapes, low volume), thread rolling (cold work, better fatigue).
- Heat treatment: annealing (soft), normalizing, quench and temper, case hardening (carburizing or nitriding).

2. Essential formulas

Name	Formula	Variables / units
Spindle speed	$N = 12V/(\pi D)$	
Cutting time	$t = L/(fN)$	
Taylor	$VT^n = C$	
MRR (turning)	$MRR = 12V \cdot f \cdot d$	in^3/min

3. Handbook search keywords

cutting speed | Taylor tool life | machining time | manufacturing processes

4. Worked examples

Example 1: Taylor tool life
n = 0.25, C = 400. Find T at V = 300 ft/min.

1. T = (400/300)^(1/0.25) = 3.16 min.

Answer: 3.16 min

5. Common traps

- Leaving out the 12 in N = 12V/πD.
- Inverting the exponent in Taylor (use 1/n).

6.A Manufacturing Methods: one-page cheat summary

► $N = 12V/\pi D$

► $t = L/fN$

► $T = (C/V)^{(1/n)}$

Formulas

Spindle speed	$N = 12V/(\pi D)$
Cutting time	$t = L/(fN)$
Taylor	$VT^n = C$
MRR (turning)	$MRR = 12V \cdot f \cdot d$

Search keywords

cutting speed, Taylor tool life, machining time, manufacturing processes

Top traps

- Leaving out the 12 in N = 12V/πD.
- Inverting the exponent in Taylor (use 1/n).

6.B Fits and Tolerances

1. Key concepts

- Hole-basis system: the hole lower limit equals basic size (H). Max clearance = largest hole − smallest shaft. Min clearance (or max interference) = smallest hole − largest shaft.
- ANSI B4.1 classes: RC running, LC locational clearance, LT transition, LN locational interference, FN force/shrink. ISO: H7/g6 sliding, H7/h6 locational, H7/p6 interference, H7/s6 drive.
- Tolerance stack: worst case $T = \Sigma T_i$; statistical (RSS) $T = \sqrt{\Sigma T_i^2}$.

2. Essential formulas

Name	Formula	Variables / units
Max clearance	$C_{max} = hole_{max} - shaft_{min}$	
Min clearance	$C_{min} = hole_{min} - shaft_{max}$ (negative = interference)	
Worst case	$T = \Sigma t_i$	
RSS	$T = \sqrt{\Sigma t_i^2}$	

3. Handbook search keywords

limits and fits · ANSI B4.1 · ISO 286 · tolerance stack-up RSS

4. Worked examples

Example 1: Stack-up
Four parts at ±0.002, ±0.003, ±0.002, ±0.004 in. Find the worst-case and RSS tolerances.

1. Worst case = ±0.011 in.
2. RSS = $\sqrt{(4 + 9 + 4 + 16)} \times 10^{-3}$ = ±0.0057 in.

Answer: ±0.011 (WC), ±0.0057 (RSS)

5. Common traps

- Swapping max and min clearance.
- Using RSS when the question asks for worst case.

6.B Fits and Tolerances: one-page cheat summary

- ▶ $C_{max} = H_{max} - s_{min}$
- ▶ $C_{min} = H_{min} - s_{max}$
- ▶ WC Σt; RSS $\sqrt{\Sigma t^2}$
- ▶ FN = force fit

Formulas

Max clearance	$C_{max} = \text{hole}_{max} - \text{shaft}_{min}$
Min clearance	$C_{min} = \text{hole}_{min} - \text{shaft}_{max}$ (negative = interference)
Worst case	$T = \Sigma t_i$
RSS	$T = \sqrt{\Sigma t_i^2}$

Search keywords

limits and fits, ANSI B4.1, ISO 286, tolerance stack-up RSS

Top traps

- Swapping max and min clearance.
- Using RSS when the question asks for worst case.

6.C Codes and Standards

1. Key concepts

- Know which code covers what:
- ASME BPVC Section VIII: unfired pressure vessels. Section I: power boilers. ASME B31.1: power piping. B31.3: process piping.
- ASME Y14.5: GD&T. Y14.100: drawing practice. AWS A2.4: weld symbols. AWS D1.1: structural steel welding.
- AGMA 2001: gear rating. ANSI/ABMA 9/11 and ISO 281: bearing ratings. SAE J429 and ASTM A325/A490: bolt grades. ASTM E8: tension test. ISO 286 and ANSI B4.1: fits.
- The exam may supply excerpts of specific standards. Check the last page of the NCEES MDM specification for the current list.

2. Essential formulas

Name	Formula	Variables / units
(no formulas)	Recognition topic. Know the scope of each code.	

3. Handbook search keywords

ASME Section VIII | B31.3 | Y14.5 | AWS D1.1 | AGMA | ABMA

4. Worked examples

Example 1: Identify the code
You are designing process piping in a refinery. Which code governs?

1. Process piping falls under ASME B31.3 (power-plant piping is B31.1).

Answer: ASME B31.3

5. Common traps

- Confusing B31.1 (power) with B31.3 (process).
- Confusing AWS A2.4 (symbols) with D1.1 (structural welding).

6.C Codes and Standards: one-page cheat summary

- ▶ BPVC VIII vessels
- ▶ B31.1 power / B31.3 process
- ▶ Y14.5 GD&T
- ▶ AWS D1.1 welding
- ▶ AGMA gears; ABMA bearings
- ▶ SAE J429 bolts

Formulas

(no formulas)	Recognition topic. Know the scope of each code.

Search keywords

ASME Section VIII, B31.3, Y14.5, AWS D1.1, AGMA, ABMA

Top traps

- Confusing B31.1 (power) with B31.3 (process).
- Confusing AWS A2.4 (symbols) with D1.1 (structural welding).

6.D Computational Methods

1. Key concepts

- Linear interpolation from property tables: $y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$.
- Root finding: Newton–Raphson $x_1 = x_0 - f(x_0)/f'(x_0)$. Bisection is slower but always converges.
- Numerical integration: trapezoid; Simpson's 1/3 rule (even number of intervals) is exact for cubics.
- FEA: refine the mesh where gradients are high. Sharp re-entrant corners are singularities, so model the real fillet. Check reactions against the hand-calculated loads.

2. Essential formulas

Name	Formula	Variables / units
Interpolation	$y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$	
Newton–Raphson	$x_{k+1} = x_k - f/f'$	
Simpson 1/3	$\int \approx (h/3)[f_0 + 4f_1 + 2f_2 + \ldots + f_n]$	
Trapezoid	$\int \approx (h/2)[f_0 + 2f_1 + \ldots + f_n]$	

3. Handbook search keywords

linear interpolation · Newton-Raphson · Simpson's rule · finite element

4. Worked examples

Example 1: Newton–Raphson
Find the root of $f(x) = x^3 - 2x - 5$ starting at $x_0 = 2$ (one iteration).

1. $f(2) = -1$; $f'(x) = 3x^2 - 2 = 10$.
2. $x_1 = 2 - (-1)/10 = 2.1$.

Answer: 2.1 (true root ≈ 2.0946)

5. Common traps

- Using Simpson's rule with an odd number of intervals.
- Inverting the slope in Newton's method.

6.D Computational Methods: one-page cheat summary

► Interp: $y_1 + \Delta y \cdot (x - x_1)/\Delta x$
► N–R: $x - f/f'$
► Simpson: $h/3[1\ 4\ 2\ 4\ \ldots 1]$
► FEA: singular corners do not converge

Formulas

Interpolation	$y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$
Newton–Raphson	$x_{k+1} = x_k - f/f'$
Simpson 1/3	$\int \approx (h/3)[f_0 + 4f_1 + 2f_2 + \ldots + f_n]$
Trapezoid	$\int \approx (h/2)[f_0 + 2f_1 + \ldots + f_n]$

Search keywords

linear interpolation, Newton-Raphson, Simpson's rule, finite element

Top traps

- Using Simpson's rule with an odd number of intervals.
- Inverting the slope in Newton's method.

6.E Instrumentation, Testing, Inspection, and Quality

1. Key concepts

- Measurement: the instrument resolution should be about 1/10 of the tolerance. A micrometer reads 0.0001 in; a caliper reads 0.001 in.
- Strain gage: ε = (ΔR/R)/GF.
- Process capability: C_p = (USL − LSL)/6σ; C_pk = min(USL − μ, μ − LSL)/3σ. 1.33 or more is good.
- Control charts: $\bar{X}$ chart limits $\bar{X} \pm A_2\bar{R}$; R chart limits $D_3\bar{R}$ and $D_4\bar{R}$.
- NDT: UT and RT find subsurface flaws. Dye penetrant finds surface-breaking flaws only. Magnetic particle finds surface and near-surface flaws (ferromagnetic parts only).

2. Essential formulas

Name	Formula	Variables / units
Strain gage	$\varepsilon = (\Delta R/R)/GF$	
Cp	$C_p = (USL - LSL)/(6\sigma)$	
Cpk	$C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$	
$\bar{X}$ chart	$UCL = \bar{\bar{X}} + A_2\bar{R}$	

3. Handbook search keywords

process capability Cpk · control chart · strain gage factor · nondestructive testing

4. Worked examples

Example 1: Cpk
Limits 25.00–25.10 mm, μ = 25.035, σ = 0.010. Find C_p and C_pk.

1. C_p = 0.10/0.06 = 1.67.
2. C_pk = min(0.065, 0.035)/0.03 = 1.17 (not centered).

Answer: C_p = 1.67, C_pk = 1.17

5. Common traps

- Using C_p when the process is off center.
- Expecting dye penetrant to find internal flaws.

6.E Instrumentation, Testing, Inspection, and Quality: one-page cheat summary

- ► $\varepsilon = (\Delta R/R)/GF$
- ► $C_{pk} = \min(\ldots)/3\sigma$
- ► $UCL = \bar{\bar{X}} + A_2\bar{R}$
- ► UT/RT subsurface; PT surface only

Formulas

Strain gage	$\varepsilon = (\Delta R/R)/GF$
Cp	$C_p = (USL - LSL)/(6\sigma)$
Cpk	$C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$
$\bar{X}$ chart	$UCL = \bar{\bar{X}} + A_2\bar{R}$

Search keywords

process capability Cpk, control chart, strain gage factor, nondestructive testing

Top traps

- Using C_p when the process is off center.
- Expecting dye penetrant to find internal flaws.

6.F Chemical Processes (Corrosion, Coatings, Treatments)

1. Key concepts

- Galvanic corrosion: the more anodic metal corrodes (Mg, Zn, Al, steel, … Cu, stainless (passive), Ti, graphite). Avoid a small anode with a large cathode; isolate dissimilar metals.
- Protection: coatings, galvanizing (zinc as a sacrificial anode), cathodic protection, inhibitors, material selection.
- Hydrogen embrittlement: high-strength steels after plating or pickling. Bake after plating.
- Surface treatments: carburizing and nitriding (hard case), shot peening (compressive residual stress), anodizing (aluminum).

2. Essential formulas

Name	Formula	Variables / units
(mostly concept)	Galvanic series order; area ratio effect	

3. Handbook search keywords

galvanic series | corrosion protection | hydrogen embrittlement | surface treatment

4. Worked examples

Example 1: Galvanic pair
Aluminum rivets in a stainless panel in seawater. What happens and how do you fix it?

1. Aluminum is anodic and the rivet area is small: severe rivet corrosion.
2. Fix: use stainless rivets (large anode, small cathode), isolate the metals, or coat the cathode.

Answer: The rivets corrode; change the material or isolate

5. Common traps

- Coating only the anode (any holiday then corrodes fast). Coat the cathode, or both.

6.F Chemical Processes (Corrosion, Coatings, Treatments): one-page cheat summary

- ► Anode corrodes
- ► Small anode/large cathode = bad
- ► Zinc protects steel sacrificially
- ► Bake after plating

Formulas

(mostly concept)	Galvanic series order; area ratio effect

Search keywords

galvanic series, corrosion protection, hydrogen embrittlement, surface treatment

Top traps

- Coating only the anode (any holiday then corrodes fast). Coat the cathode, or both.